

10 CSTRs

Why are we learning this?

In previous lectures, we developed the tools needed for reactor design:

- Mole balances
- Conversion
- Rate laws
- Equilibrium
- Concentration expressions

Today we will apply those tools to one of the most important reactor types in chemical engineering: the **Continuous Stirred Tank Reactor (CSTR)**. By the end of today's lesson, you should understand:

- The defining assumptions of a CSTR.
- Why perfect mixing simplifies reactor analysis.
- How to interpret the CSTR mole balance.
- How conversion affects required reactor volume.
- How reaction rate affects required reactor volume.
- How to solve reactor design problems involving CSTRs.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress
(Conversion)



Graphical Reactor Design

(Inverse Rate/Levenspiel Plots)



Rate Laws

(Kinetics)



Equilibrium

(Reversible Reactions)



Equilibrium Conversion

(Thermodynamic Limitations)



One Reaction: Solving in terms of Reaction Progress



► One Reaction: Reactor Design ◀

(CSTR, Batch, PFR, PBR)



Catalytic Reactions



Numerical Methods

(Python)



Energy Balances



One Reaction: Non-Isothermal Reactor Design



Multiple Reactions



Reactor Safety



Challenging the Well-Mixed Assumption

(Residence Time Distributions)



Challenging the Elementary Rate Law Assumption

(Reaction Mechanisms)

Big Idea

A CSTR design problem usually asks: How large must a single well-mixed volume be to achieve a desired conversion?

Today we learn how to answer that question.

A key characteristic of the CSTR is

Specifically, CSTRs operate at

Hence, the entire volume operates at

This phenomenon requires ; we will show later
in the class that this is

The Defining Characteristic of a CSTR

A CSTR is assumed to be **perfectly mixed**. Perfect mixing means:

Every location inside the reactor has the same composition.

As a result:

Reactor conditions = Outlet conditions

This assumption is what makes the CSTR easy to analyze.

A Big Consequence of Perfect Mixing

Because every part of the reactor has the same composition:

Every part of the reactor operates at the outlet rate.

This means the reaction rate used in the design equation is evaluated using outlet conditions. This is one of the most important ideas in CSTR design.

The Cost of Perfect Mixing

Perfect mixing greatly simplifies analysis. However, it also means that the entire reactor operates at the slowest reaction conditions found in the reactor: the outlet conditions. As a result:

CSTRs are often less volume-efficient than PFRs.

We will see this more clearly when comparing reactor types.

CSTR mole balance:

- In terms of number of moles (N): $V = \frac{F_{A0} - F_A}{-r_A}$. In words, the reactor volume is equal to the amount of species A reacted in the reactor divided by the rate of consumption of species A.
 - If species A is a reactant,
 - If species A is a product,

- If the "amount reacted" is larger, the volume needed is
- If the rate of consumption is larger, the volume needed is

What Does the CSTR Mole Balance Mean?

The CSTR design equation is:

Reactor Volume = Amount Reacted ÷ Reaction Rate

This equation has a clear physical interpretation: To achieve a larger amount of reaction, we generally need a larger reactor. If a reaction has a faster reaction rate, it generally requires a smaller reactor.

- In terms of conversion (X): $V = \frac{F_{A0}X_A}{-r_A}$
- Same as with batch reactors, if you are solving a reactor design problem for a prototypical CSTR, you can just write down the CSTR mole balance without re-deriving it each time, with the following exceptions.
 - The problem statement asks you to derive the balance.
 - The reactor is not a prototypical CSTR.

Use the Design Equation Efficiently

For a standard CSTR problem, the reactor design equation is already known. Unless:

- the problem explicitly requests a derivation, or
- the reactor does not satisfy CSTR assumptions,

you can begin directly with the CSTR design equation. Engineering calculations often focus on using models appropriately rather than repeatedly deriving them.

- Let's do an example: The gas phase reaction $2A \leftrightarrow B$ exhibits elementary kinetics. It is carried out in a CSTR to 95% of the equilibrium conversion with $P_{A0} = 10$ atm and $T = 400$ K. $K_C = 40$ L/mol, $k_f = 0.1$ L/mol/s, and $v_0 = 1$ L/s. Determine the volume of the CSTR.

Think Before Calculating

Before beginning a reactor design calculation, ask: What quantity am I solving for?

For a CSTR, that quantity is usually: Required reactor volume

Then identify:

- The appropriate form of the mole balance.
- The desired conversion.
- The rate law.
- The concentrations needed to evaluate the rate.
- Any equilibrium limitations.

This approach prevents many common mistakes.

Summary and Key Takeaways

Today we developed the reactor design framework for Continuous Stirred Tank Reactors (CSTRs).

Key Ideas

- ✓ A CSTR is assumed to be perfectly mixed.
- ✓ Reactor conditions are identical to outlet conditions.
- ✓ The reaction rate used in the design equation is evaluated at outlet conditions.
- ✓ Larger desired conversion generally requires larger reactor volume.
- ✓ Faster reaction rates generally reduce required reactor volume.
- ✓ For standard reactor problems, the established CSTR design equation can be used directly.

Most Important Idea

Perfect Mixing



Uniform Reactor Conditions



Simple Design Equation

The simplicity of the CSTR comes directly from the assumption of perfect mixing. That same assumption also influences reactor performance and required volume.